

# Fixed Points of a Random Permutation

Reproduction and Revised Editions

**Source.** This document is an independent LaTeX reproduction of, and editorial adaptation based on, the mathematical paper shown in the Bilibili video **【日常研究】摆烂小能手掀起的数学问题** (video ID BV1Fh411N71S):

<https://www.bilibili.com/video/BV1Fh411N71S>

**Repository.** The  $\text{\LaTeX}$  sources, compiled PDFs, and edition notes are published at:

<https://github.com/Ritsuka314/personal-website>

**PDF editions.**

`paper-en.pdf`   `paper-en-revised.pdf`   `paper-zh.pdf`

**Purpose.** The reconstruction was prepared solely for entertainment and educational purposes. It is unofficial, is not presented as an author-approved transcript, and makes no claim of affiliation with the video creator or platform.

**The three editions.**

1. **Reproduced English edition.** A close typesetting of the four-page English note visible in the video. It intentionally retains the source's nonstandard wording, omissions, notation, and induction argument.
2. **Revised English edition.** A corrected, self-contained mathematical report. It defines the model precisely, proves the derangement formula by inclusion–exclusion, gives a unified probability distribution, and proves the expectation with indicator variables.
3. **Revised Chinese edition.** A polished Chinese counterpart to the revised English report. It follows the same definitions, results, and proofs rather than translating the reproduced wording literally.

**Edition in this file: Revised English edition**

This edition is a corrected, self-contained report based on the mathematical question in the source video.

# Fixed Points of a Random Permutation

Distribution and Expected Value

## Abstract

Let a permutation of  $n$  distinct objects be chosen uniformly at random, and let  $X$  denote its number of fixed points. This report derives the exact distribution of  $X$  by counting derangements and proves that  $\mathbb{E}[X] = 1$  for every positive integer  $n$ . The result models, for example, the number of correct answers obtained when  $n$  distinct choices are assigned uniformly at random to  $n$  distinct blanks.

## 1 Problem setup

For a positive integer  $n$ , write

$$[n] = \{1, 2, \dots, n\}.$$

Let  $\sigma : [n] \rightarrow [n]$  be a permutation chosen uniformly from the  $n!$  elements of the symmetric group  $S_n$ . An index  $r \in [n]$  is a *fixed point* of  $\sigma$  if  $\sigma(r) = r$ . Define

$$X = |\{r \in [n] : \sigma(r) = r\}|.$$

Our goals are to determine the probability mass function of  $X$  and its expected value.

## 2 Counting derangements

A *derangement* is a permutation with no fixed points. Let  $D_m$  denote the number of derangements of  $m$  objects, with the convention  $D_0 = 1$ .

**Theorem 1** (Derangement formula). *For every integer  $m \geq 0$ ,*

$$D_m = m! \sum_{j=0}^m \frac{(-1)^j}{j!}.$$

*Consequently, a uniformly random permutation of  $m$  objects is a derangement with probability*

$$\frac{D_m}{m!} = \sum_{j=0}^m \frac{(-1)^j}{j!}.$$

*Proof.* For  $r \in [m]$ , let  $A_r$  be the set of permutations that fix  $r$ . If a set  $J \subseteq [m]$  has  $j$  elements,

then exactly  $(m - j)!$  permutations fix every element of  $J$ . Inclusion–exclusion therefore gives

$$\begin{aligned} D_m &= \left| \bigcap_{r=1}^m A_r^c \right| \\ &= \sum_{j=0}^m (-1)^j \binom{m}{j} (m - j)! \\ &= m! \sum_{j=0}^m \frac{(-1)^j}{j!}. \end{aligned}$$

Dividing by the total number  $m!$  of permutations proves the probability formula.  $\square$

When  $m \geq 2$ , the terms for  $j = 0$  and  $j = 1$  cancel. Thus the same probability may also be written as

$$\sum_{j=2}^m \frac{(-1)^j}{j!},$$

which is the form used in the reproduced note.

### 3 Distribution of the number of fixed points

**Theorem 2.** For  $k = 0, 1, \dots, n$ ,

$$\Pr(X = k) = \frac{\binom{n}{k} D_{n-k}}{n!} = \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}.$$

*Proof.* First choose the  $k$  indices that will be fixed; this can be done in  $\binom{n}{k}$  ways. The remaining  $n - k$  indices must then be deranged, which can be done in  $D_{n-k}$  ways. Hence exactly  $\binom{n}{k} D_{n-k}$  permutations have  $k$  fixed points. Dividing by  $n!$  and substituting the derangement formula yields

$$\begin{aligned} \Pr(X = k) &= \frac{\binom{n}{k} D_{n-k}}{n!} \\ &= \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}. \end{aligned}$$

$\square$

The unified formula includes the two boundary cases:

$$\Pr(X = n) = \frac{1}{n!}, \quad \Pr(X = n - 1) = 0.$$

The second identity holds because a permutation cannot move exactly one element while fixing every other element.

## 4 Expected number of fixed points

**Theorem 3.** For every  $n \geq 1$ ,

$$\mathbb{E}[X] = 1.$$

*Proof.* For each  $r \in [n]$ , define the indicator variable

$$I_r = \begin{cases} 1, & \sigma(r) = r, \\ 0, & \sigma(r) \neq r. \end{cases}$$

Then  $X = \sum_{r=1}^n I_r$ . Of the  $n!$  permutations, exactly  $(n-1)!$  fix a specified index  $r$ , so

$$\Pr(\sigma(r) = r) = \frac{(n-1)!}{n!} = \frac{1}{n} \quad \text{and} \quad \mathbb{E}[I_r] = \frac{1}{n}.$$

By linearity of expectation,

$$\mathbb{E}[X] = \sum_{r=1}^n \mathbb{E}[I_r] = n \cdot \frac{1}{n} = 1.$$

The indicator variables need not be independent for this argument. □

## 5 Conclusion

The number of fixed points in a uniformly random permutation has the exact distribution

$$\Pr(X = k) = \frac{\binom{n}{k} D_{n-k}}{n!},$$

and its mean is always one. Therefore, when  $n$  distinct answers are placed uniformly at random into  $n$  distinct blanks, one answer is correct on average. This is an expectation statement: an individual attempt may have no correct answers, one correct answer, or several.